

Statistical and Machine Learning SS26

Solutions: Exercise Sheet 2

Exercise 1: Melbourne

a)

```
House <- read.csv("Melbourne_housing_FULL.csv")
str(House) #number of observations, variables, types of variables
```

```
## 'data.frame': 34857 obs. of 21 variables:
## $ Suburb : chr "Abbotsford" "Abbotsford" "Abbotsford" "Abbotsford" ...
## $ Address : chr "68 Studley St" "85 Turner St" "25 Bloomburg St" "18/659 Victoria St" ...
## $ Rooms : int 2 2 2 3 3 3 4 4 2 2 ...
## $ Type : chr "h" "h" "h" "u" ...
## $ Price : int NA 1480000 1035000 NA 1465000 850000 1600000 NA NA NA ...
## $ Method : chr "SS" "S" "S" "VB" ...
## $ SellerG : chr "Jellis" "Biggin" "Biggin" "Rounds" ...
## $ Date : chr "3/09/2016" "3/12/2016" "4/02/2016" "4/02/2016" ...
## $ Distance : num 2.5 2.5 2.5 2.5 2.5 2.5 2.5 2.5 2.5 2.5 ...
## $ Postcode : int 3067 3067 3067 3067 3067 3067 3067 3067 3067 3067 ...
## $ Bedroom2 : int 2 2 2 3 3 3 3 3 4 3 ...
## $ Bathroom : int 1 1 1 2 2 2 1 2 1 2 ...
## $ Car : int 1 1 0 1 0 1 2 2 2 1 ...
## $ Landsize : int 126 202 156 0 134 94 120 400 201 202 ...
## $ BuildingArea : num NA NA 79 NA 150 NA 142 220 NA NA ...
## $ YearBuilt : int NA NA 1900 NA 1900 NA 2014 2006 1900 1900 ...
## $ CouncilArea : chr "Yarra City Council" "Yarra City Council" "Yarra City Council" "Yarra City Council" ...
## $ Lattitude : num -37.8 -37.8 -37.8 -37.8 -37.8 ...
## $ Longitude : num 145 145 145 145 145 ...
## $ Regionname : chr "Northern Metropolitan" "Northern Metropolitan" "Northern Metropolitan" "Northern Metropolitan" ...
## $ Propertycount: int 4019 4019 4019 4019 4019 4019 4019 4019 4019 4019 ...
```

```
summary(House) #min/max values, number of missing values
```

```
## Suburb Address Rooms Type
## Length:34857 Length:34857 Min. : 1.000 Length:34857
## Class :character Class :character 1st Qu.: 2.000 Class :character
## Mode :character Mode :character Median : 3.000 Mode :character
## Mean : 3.031
## 3rd Qu.: 4.000
## Max. :16.000
##
## Price Method SellerG Date
## Min. : 85000 Length:34857 Length:34857 Length:34857
```

```
## 1st Qu.: 635000 Class :character Class :character Class :character
## Median : 870000 Mode :character Mode :character Mode :character
## Mean : 1050173
## 3rd Qu.: 1295000
## Max. :11200000
## NA's :7610
## Distance Postcode Bedroom2 Bathroom
## Min. : 0.00 Min. :3000 Min. : 0.000 Min. : 0.000
## 1st Qu.: 6.40 1st Qu.:3051 1st Qu.: 2.000 1st Qu.: 1.000
## Median :10.30 Median :3103 Median : 3.000 Median : 2.000
## Mean :11.18 Mean :3116 Mean : 3.085 Mean : 1.625
## 3rd Qu.:14.00 3rd Qu.:3156 3rd Qu.: 4.000 3rd Qu.: 2.000
## Max. :48.10 Max. :3978 Max. :30.000 Max. :12.000
## NA's :1 NA's :1 NA's :8217 NA's :8226
## Car Landsize BuildingArea YearBuilt
## Min. : 0.000 Min. : 0.0 Min. : 0.0 Min. :1196
## 1st Qu.: 1.000 1st Qu.: 224.0 1st Qu.: 102.0 1st Qu.:1940
## Median : 2.000 Median : 521.0 Median : 136.0 Median :1970
## Mean : 1.729 Mean : 593.6 Mean : 160.3 Mean :1965
## 3rd Qu.: 2.000 3rd Qu.: 670.0 3rd Qu.: 188.0 3rd Qu.:2000
## Max. :26.000 Max. :433014.0 Max. :44515.0 Max. :2106
## NA's :8728 NA's :11810 NA's :21115 NA's :19306
## CouncilArea Latitude Longitude Regionname
## Length:34857 Min. : -38.19 Min. :144.4 Length:34857
## Class :character 1st Qu.: -37.86 1st Qu.:144.9 Class :character
## Mode :character Median : -37.81 Median :145.0 Mode :character
## Mean : -37.81 Mean :145.0
## 3rd Qu.: -37.75 3rd Qu.:145.1
## Max. : -37.39 Max. :145.5
## NA's :7976 NA's :7976
## Propertycount
## Min. : 83
## 1st Qu.: 4385
## Median : 6763
## Mean : 7573
## 3rd Qu.:10412
## Max. :21650
## NA's :3
```

b)

```
index <- House$Suburb %in% c("Brunswick", "Craigieburn", "Hawthorn")
# %in% checks for each row whether Suburb is contained
# in the vector of three suburb names. Returns a logical vector of
# TRUE/FALSE with the same length as the number of rows in House
House_sub <- House[index,] #Only rows where index == TRUE are kept. The comma with nothing
# after it means keep all columns
```

c)

```
str(House_sub) # overview of sub data set
```

```
## 'data.frame': 1127 obs. of 21 variables:
```

```
## $ Suburb      : chr "Brunswick" "Brunswick" "Brunswick" "Brunswick" ...
## $ Address     : chr "8/55 De Carle St" "234 Victoria St" "9 Blair St" "25 Hardy St" ...
## $ Rooms       : int 2 2 4 4 3 2 2 4 4 3 ...
## $ Type        : chr "u" "h" "h" "h" ...
## $ Price       : int 428000 1026000 1802500 1500000 NA 665000 570000 1330000 2230000 1050000 ...
## $ Method      : chr "S" "S" "S" "VB" ...
## $ SellerG     : chr "Brad" "Nelson" "Jellis" "Nelson" ...
## $ Date        : chr "3/09/2016" "3/09/2016" "3/12/2016" "3/12/2016" ...
## $ Distance    : num 5.2 5.2 5.2 5.2 5.2 5.2 5.2 5.2 5.2 5.2 ...
## $ Postcode    : int 3056 3056 3056 3056 3056 3056 3056 3056 3056 3056 ...
## $ Bedroom2    : int 2 2 4 4 NA NA NA NA 4 3 ...
## $ Bathroom    : int 1 1 3 3 NA NA NA NA 2 1 ...
## $ Car         : int 2 0 1 1 NA NA NA NA 3 2 ...
## $ Landsize    : int 0 225 238 365 NA NA NA NA 851 264 ...
## $ BuildingArea : num NA 108 93 209 NA NA NA NA 287 NA ...
## $ YearBuilt   : int 1970 1905 1910 1890 NA NA NA NA 1940 NA ...
## $ CouncilArea : chr "Moreland City Council" "Moreland City Council" "Moreland City Council" "Moreland City Council" ...
## $ Lattitude   : num -37.8 -37.8 -37.8 -37.8 NA ...
## $ Longitude   : num 145 145 145 145 NA ...
## $ Regionname  : chr "Northern Metropolitan" "Northern Metropolitan" "Northern Metropolitan" "Northern Metropolitan" ...
## $ Propertycount : int 11918 11918 11918 11918 11918 11918 11918 11918 11918 11918 ...
```

```
Price <- as.numeric(House$Price) #convert variables to numeric
# (if they would've been character)
House_nmissing <- na.omit(House_sub) #removes observations with missing values
summary(House_nmissing$Price)
```

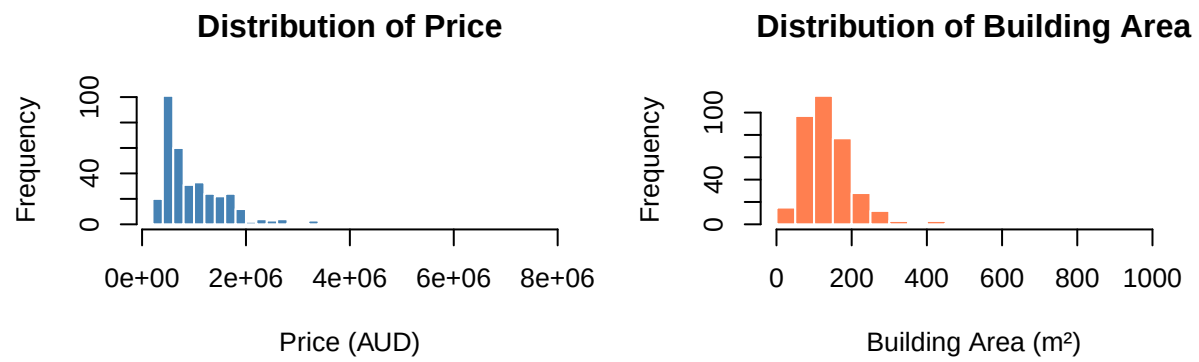
```
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 272500  558000  765000 1063723 1342500 7650000
```

```
summary(House_nmissing$BuildingArea)
```

```
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      0.0   89.0   124.5   137.6   171.0  1041.0
```

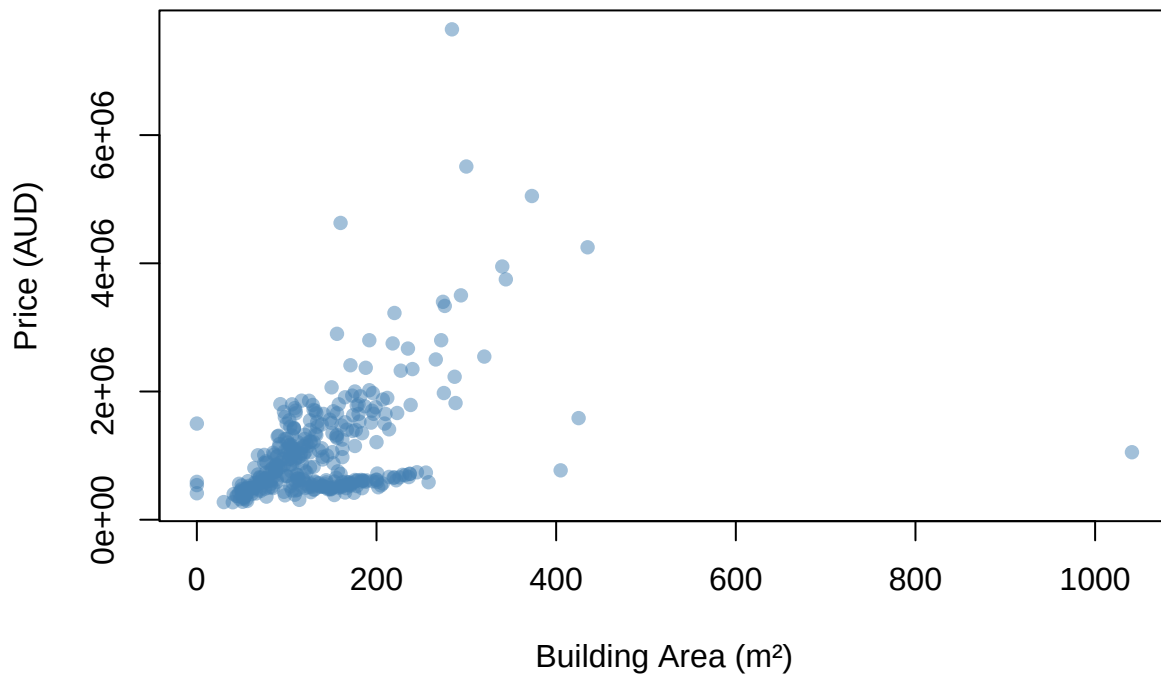
```
par(mfrow = c(2, 2))
hist(House_nmissing$Price,
     breaks = 30,
     main   = "Distribution of Price",
     xlab   = "Price (AUD)",
     col    = "steelblue",
     border = "white")

hist(House_nmissing$BuildingArea,
     breaks = 30,
     main   = "Distribution of Building Area",
     xlab   = "Building Area (m²)",
     col    = "coral",
     border = "white")
```



```
plot(House_nmissing$BuildingArea, House_nmissing$Price,
     main = "Price vs Building Area",
     xlab = "Building Area (m²)",
     ylab = "Price (AUD)",
     pch = 16,
     col = adjustcolor("steelblue", alpha.f = 0.5))
```

Price vs Building Area



Price: right-skewed distribution, most houses cost between 0 and 2 Mio AUD, price of the few outliers could influence slope of regression
 BuildingArea: also right-skewed, most houses have areas between 0 and 300 sqm. Same thing here: outliers could influence slope of regression

d)

```
fit_simple <- lm(Price ~ BuildingArea, data = House_nmissing)
summary(fit_simple)
```

```
##
## Call:
## lm(formula = Price ~ BuildingArea, data = House_nmissing)
##
## Residuals:
```

	Min	1Q	Median	3Q	Max
	-4187963	-435096	-92670	263547	5909383

```
##
## Coefficients:
```

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	427409.1	76905.2	5.558	5.43e-08 ***
BuildingArea	4624.0	479.6	9.640	< 2e-16 ***

```
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 740500 on 350 degrees of freedom
## Multiple R-squared:  0.2098, Adjusted R-squared:  0.2076
```

```
## F-statistic: 92.94 on 1 and 350 DF, p-value: < 2.2e-16
```

Intercept: On average, a house in with 0 sqm costs AUD 427,409.1. With every added sqm, the price increases by approximately AUD 4,624. This effect is statistically significant on a 1% level (p value < 0.01)

```
fit_multiple <- lm(Price~ BuildingArea + Suburb, data = House_nmissing)
summary(fit_multiple)
```

```
##
## Call:
## lm(formula = Price ~ BuildingArea + Suburb, data = House_nmissing)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -4927775  -257168   -20874   192805  5328917
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    382976.5     72517.8   5.281 2.27e-07 ***
## BuildingArea      5377.3       405.1  13.274 < 2e-16 ***
## SuburbCraigieburn -685235.2     81883.5  -8.368 1.44e-15 ***
## SuburbHawthorn    410945.5     78072.2   5.264 2.48e-07 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 617000 on 348 degrees of freedom
## Multiple R-squared:  0.4546, Adjusted R-squared:  0.4499
## F-statistic: 96.69 on 3 and 348 DF, p-value: < 2.2e-16
```

Intercept: On average, a house with 0 sqm in Brunswick costs AUD 382976.5 . With every added sqm, the price increases by approximately AUD 5,377, holding suburb constant. Houses in Craigieburn cost approximately AUD 685,235 less than those in Brunswick, holding sqm constant. For a house in Hawthorn on the other hand one would have to pay around AUD 410,945 more, compared to those in Brunswick, cp. All coefficients are statistically significant on a 1% level (p value < 0.01) The model improves because the adjusted R^2 increases (0.2076 to 0.4499). Unlike regular R^2 , adjusted R^2 penalizes for adding extra predictors. If it goes up, the new predictor genuinely helps.

- e) Outliers: observation with large residuals (large error terms; its actual value is far from what the model predicts. It is surprising in terms of y), not necessarily changes the model

Influential observations: an observation whose removal would substantially change the regression coefficients. It is surprising in terms of x (high leverage) and pulls the regression line toward itself.

```
cd <- cooks.distance(fit_multiple) #Computes cd for every observation and stores it as a
# numeric vector of length n.
#we will solve it manually in the next exercise

# Suspected outliers (small or large building area)
outlier_idx <- which(House_nmissing$BuildingArea <= 5 |
                    House_nmissing$BuildingArea > 300)

cbind(House_nmissing[outlier_idx, c("Suburb", "BuildingArea", "Price")],
      CooksD = round(cd[outlier_idx], 4))
```

```
##           Suburb BuildingArea   Price   CooksD
## 2297    Brunswick         1041 1053000 14.4120
## 5277    Hawthorn          340 3950000  0.0350
## 5284    Hawthorn          435 4250000  0.0451
## 5313    Hawthorn          373 5050000  0.1251
## 5352    Hawthorn          344 3750000  0.0250
## 16739   Brunswick         320 2545000  0.0030
## 23086   Craigieburn         0  412500  0.0076
## 23691   Craigieburn         0  590000  0.0118
## 26634   Craigieburn         0  540000  0.0105
## 31542   Brunswick         0 1500000  0.0116
## 32501   Brunswick         425 1585000  0.0373
## 33226   Craigieburn         405  770000  0.0316
```

```
# BuildingArea <= 5 | BuildingArea > 300 creates a TRUE/FALSE vector for every row
# which() extracts the row numbers where that condition is TRUE -> So outlier_idx is just a
# vector of indices like c(3, 17, 45, 102, ...)
# cbind then pulls the Suburb, BuildingArea and Price columns for only the suspected outlier
# rows and adds their respective Cooks distance
```

Rule of thumb: $D_\ell > 0.5 \rightarrow$ potentially influential, $D_\ell \geq 1 \rightarrow$ highly influential. In this case, we have one highly influential observation. A Cook's Distance of 14 is extremely large, this house with 1041 m² is almost certainly a data entry error or a very unusual property. We should think about removing this observation and re-fitting the model.

f)

```
fit_multiple2 <- lm(Price ~ BuildingArea + Suburb + Car, data = House_nmissing)
summary(fit_multiple2)
```

```
##
## Call:
## lm(formula = Price ~ BuildingArea + Suburb + Car, data = House_nmissing)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -4239701 -267735  -55301   258577  4900032
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    236285.6    74824.5   3.158  0.00173 **
## BuildingArea      4663.2     411.3  11.337 < 2e-16 ***
## SuburbCraigieburn -793963.6    81270.5  -9.769 < 2e-16 ***
## SuburbHawthorn    381325.7    75280.9   5.065 6.64e-07 ***
## Car             202000.4    37334.0   5.411 1.17e-07 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 593300 on 347 degrees of freedom
## Multiple R-squared:  0.497, Adjusted R-squared:  0.4912
## F-statistic: 85.73 on 4 and 347 DF, p-value: < 2.2e-16
```

```
new_house <- data.frame(BuildingArea = 100,
                        Suburb       = "Hawthorn",
                        Car           = 2)

predict(fit_multiple2, newdata = new_house, interval = "prediction", level = 0.95)
```

```
##      fit      lwr      upr
## 1 1487934 313871.4 2661997
```

The model predicts a house in Hawthorn with 100 sqm and 2 car spaces will sell for approximately AUD 1,487,934. The 95% prediction interval (AUD 313,871 - 2,661,997) means: if we were to observe many individual houses with these exact characteristics, 95% of their actual sale prices would fall within this range.

g)

```
Y <- as.numeric(House_nmissing$Price > 1e6) #create a binary outcome variable from the continuous price
# 1 = High price, 0 = Low price
House_nmissing$Y <- Y
```

Fit two classification models (LR and LDA)

```
#install.packages("tidyverse")
library("tidyverse")
```

```
## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
## v dplyr      1.2.1      v readr      2.2.0
## v forcats    1.0.1      v stringr    1.6.0
## v ggplot2    4.0.2      v tibble     3.3.1
## v lubridate  1.9.5      v tidyr      1.3.2
## v purrr      1.2.2
## -- Conflicts ----- tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()     masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

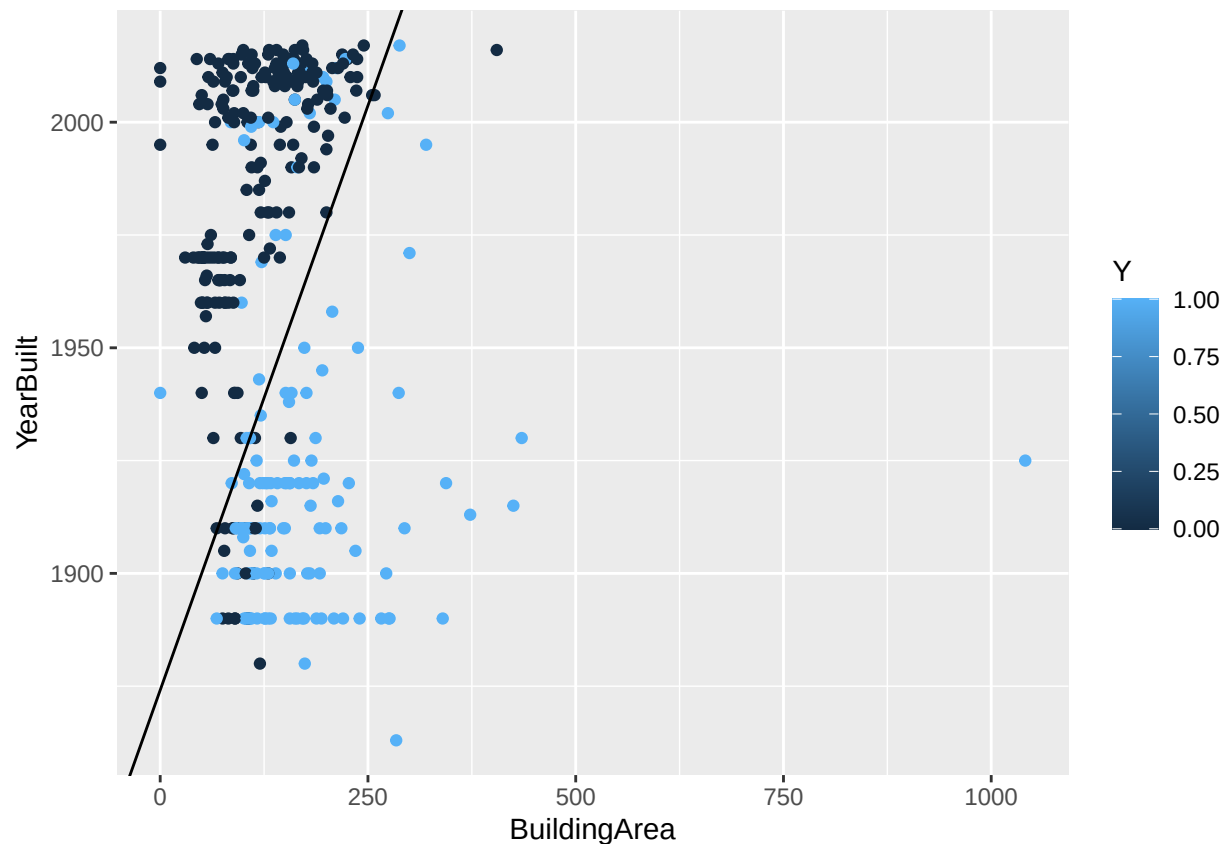
```
glm.fit <- glm(Y ~ BuildingArea + YearBuilt, data = House_nmissing, family = binomial) #LR
summary(glm.fit)
```

```
##
## Call:
## glm(formula = Y ~ BuildingArea + YearBuilt, family = binomial,
##      data = House_nmissing)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)  81.337375   8.139847   9.992 < 2e-16 ***
## BuildingArea  0.022491   0.003504   6.420 1.37e-10 ***
## YearBuilt    -0.043400   0.004299 -10.095 < 2e-16 ***
## ---
```



```
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##    Null deviance: 473.14  on 351  degrees of freedom
## Residual deviance: 254.74  on 349  degrees of freedom
## AIC: 260.74
##
## Number of Fisher Scoring iterations: 6
```

```
ggplot(House_nmissing) +
  geom_point(aes(x = BuildingArea, y = YearBuilt, color = Y)) +
  geom_abline(
    slope = -glm.fit$coefficients[2] / glm.fit$coefficients[3],
    intercept = -glm.fit$coefficients[1] / glm.fit$coefficients[3]
  )
```



```
library(MASS)
```

```
##
## Attaching package: 'MASS'

## The following object is masked from 'package:dplyr':
##
##    select
```

```

model <- lda(Y ~ BuildingArea + YearBuilt, data = House_nmissing) #LDA

#plot:
library(tidyverse)
library(grid) # for arrow()

w <- model$scaling[, 1] # discriminant direction
means <- model$means
mid <- colMeans(means) # midpoint between class means

# decision boundary: perpendicular to w, passing through midpoint
slope <- -w[1] / w[2]
intercept <- mid[2] - slope * mid[1]

# choose a length for the displayed direction vector
arrow_scale <- 0.6

# endpoint of direction vector starting at midpoint
x0 <- mid[1]
y0 <- mid[2]
x1 <- x0 + arrow_scale * w[1]
y1 <- y0 + arrow_scale * w[2]

ggplot(House_nmissing, aes(BuildingArea, YearBuilt, color = Y)) +
  geom_point(size = 4) +

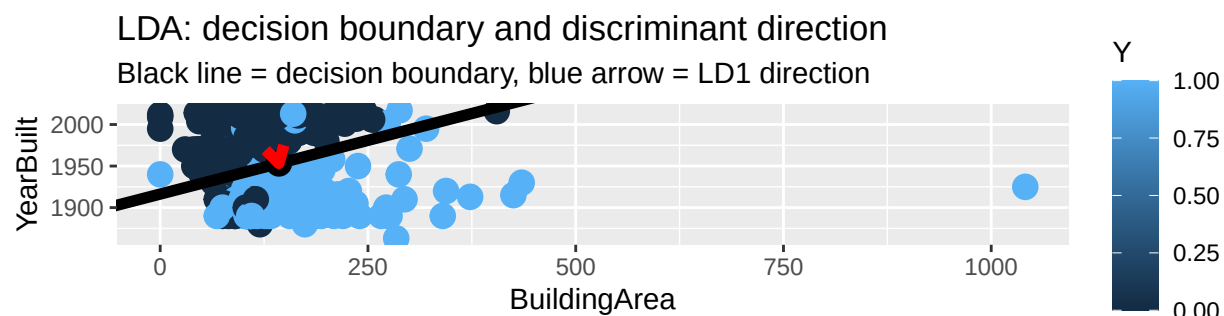
  # decision boundary
  geom_abline(slope = slope, intercept = intercept, linewidth = 2) +

  # midpoint
  annotate("point", x = x0, y = y0, size = 4, color = "black") +

  # discriminant direction vector
  annotate(
    "segment",
    x = x0, y = y0, xend = x1, yend = y1,
    colour = "red",
    linewidth = 2,
    arrow = arrow(length = unit(0.25, "cm"))
  ) +

  labs(
    title = "LDA: decision boundary and discriminant direction",
    subtitle = "Black line = decision boundary, blue arrow = LD1 direction"
  ) +
  coord_equal()

```



Exercise 2: Salaries

a)

```
salary <- read.table("salary.dat", header = TRUE)
str(salary) #sx is character, in regression R treats it as dummy variable: f=0, m=1 bc of alphabet
```

```
## 'data.frame':   52 obs. of  6 variables:
## $ sx: chr  "male" "male" "male" "female" ...
## $ rk: chr  "full" "full" "full" "full" ...
## $ yr: int   25  13  10  7  19  16  0  16  13  13 ...
## $ dg: chr  "doctorate" "doctorate" "doctorate" "doctorate" ...
## $ yd: int   35  22  23  27  30  21  32  18  30  31 ...
## $ sl: int  36350 35350 28200 26775 33696 28516 24900 31909 31850 32850 ...
```

```
salary_fit <- lm(sl ~ sx, data = salary)
summary(salary_fit)
```

```
##
## Call:
## lm(formula = sl ~ sx, data = salary)
##
## Residuals:
```

```
##      Min      1Q  Median      3Q      Max
## -8602.8 -4296.6 -100.8  3513.1 16687.9
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    21357      1545   13.820  <2e-16 ***
## sxmale         3340      1808    1.847   0.0706 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 5782 on 50 degrees of freedom
## Multiple R-squared:  0.0639, Adjusted R-squared:  0.04518
## F-statistic: 3.413 on 1 and 50 DF,  p-value: 0.0706
```

The average salary for women is \$21,357. We find that on average women make \$3,340 less than men. The difference is not significant.

- b) Salary may not only depend on sex, but also on education, age, race, level of experience -> omitted variable bias

```
salary_fit_2 <- lm(sl ~ sx + dg*yd, data = salary) #dg*yd includes all individual and interaction terms
summary(salary_fit_2)
```

```
##
## Call:
## lm(formula = sl ~ sx + dg * yd, data = salary)
##
## Residuals:
##      Min      1Q  Median      3Q      Max
## -5667.7 -1988.7 -442.5  1466.5  8810.9
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  14526.79    1220.74  11.900 8.74e-16 ***
## sxmale       1809.74    1139.37   1.588 0.118908
## dgmasters    4423.06    2703.03   1.636 0.108450
## yd           612.80     67.41    9.091 6.32e-12 ***
## dgmasters:yd -437.54     123.07  -3.555 0.000874 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 3532 on 47 degrees of freedom
## Multiple R-squared:  0.6716, Adjusted R-squared:  0.6436
## F-statistic: 24.03 on 4 and 47 DF,  p-value: 7.256e-11
```

```
salary$dg <- as.factor(salary$dg)
levels(salary$dg)
```

```
## [1] "doctorate" "masters"
```

Women who just received a PhD make \$14,527 on average, whereas those who just received a master's degree make \$4,423 more (but effect is not significant). Faculty with longer time since the degree have

higher salaries, on average \$613 per year for PhDs and \$175 (612-437) per year for masters (stat. sig. on 1% level). Female faculty make on average \$1,810 less than comparable males with the same degree and years since degree (difference not significant). The interaction term is clearly needed, with p -value < 0.01

We still see gender discrimination, even after adding control variables to the model. The discrimination is less severe (\$1810 compared to \$3340), but visible.

c) Manual computation:

```
salary <- cbind(salary, as.numeric(as.factor(salary$sx))-1,
               as.numeric(as.factor(salary$dg))-1)
names(salary)[7:8] <-c("sx.n", "dg.n")
y <- salary$s1
X <- cbind(1, salary[, c(5,7,8)], salary$yd*salary$dg.n)
n <- nrow(X); p <- ncol(X)

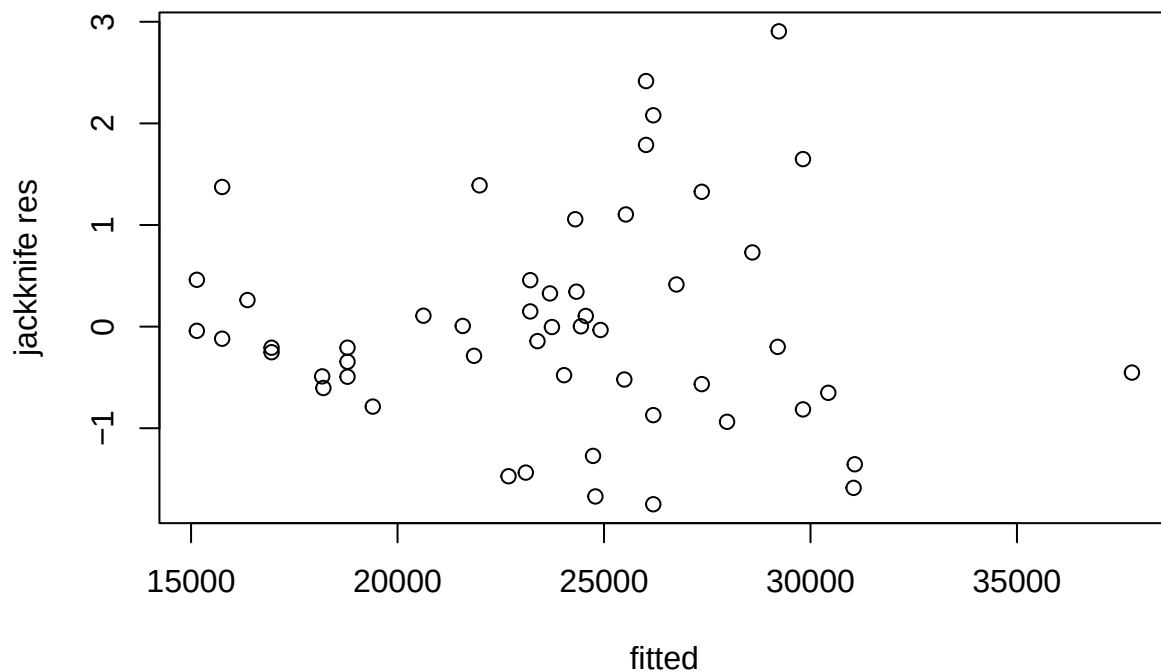
qrx <- qr(X)
qy <- qr.qty(qrx, y)
f <- qy[1:p]
r <- qy[-(1:p)]
Q <- qr.Q(qrx)
R <- qr.R(qrx)

invR <- backsolve(R, diag(ncol(R)))
sigma.hat <- sum(r^2)/(n-p)

H <- Q%*%t(Q)
fy <- H%*%y #fitted y
e <- y-fy #residuals
e <- resid(salary_fit_2)
h <- diag(H) # leverage
s <- e/sqrt(sigma.hat*(1-h)) #standardized residuals
t <- s+sqrt((n-p-1)/(n-p-s^2)) # jackknife residuals
```

alternatively:

```
t <- rstudent(salary_fit_2)
fy <- fitted(salary_fit_2)
plot(fy, t, xlab="fitted", ylab="jackknife res")
```



Plot indicates increasing variance with the mean.

List observations with large jackknife residuals:

```
salary[abs(t)>2, 1:6]
```

```
##      sx  rk yr      dg yd   sl
## 5   male full 19  masters 30 33696
## 10  male full 13  masters 31 32850
## 24 female full 8  doctorate 24 38045
```

```
2*(1-pt(abs(t)[abs(t)>2], n-p-1)) < 0.05/n #bonferroni
```

```
##      5    10    24
## FALSE FALSE FALSE
```

There are three full professors with substantially higher than expected salaries, two males and one female. They can not be formally considered as outliers when Bonferroni's correction is applied

d)

```
cookD <- s^2*h/(1-h)/p
#alt.: cookD <- cooks.distance(salary_fit_2)
cbind(salary[h>2*p/n, 1:6], cookD[h>2*p/n])
```

```
##      sx      rk yr      dg yd      sl cookD[h > 2 * p/n]
## 1  male      full 25 doctorate 35 36350      0.01089682
## 29 male associate 3   masters  7 26182      0.13023998
```

There are two observations with high leverage. Both men, one with a long duration since PhD and another with a master's degree. None of them is influential (Cook's distance < 0.5) High leverage means the observation has an unusual predictor value (extreme x), but if it still follows the regression pattern, it won't distort the coefficients much. Cook's distance captures the actual influence on fitted values.

```
i <- which.max(cookD)
c(t[i], h[i], cookD[i])
```

```
##      24      24
## 2.9057969 0.1464602 0.2501537
```

```
salary[i, 1:6]
```

```
##      sx  rk yr      dg yd      sl
## 24 female full  8 doctorate 24 38045
```

The most influential observation is the highly paid female professor with the large jackknife residual. Her Cook's distance is below one but were it not for her, our estimate of the gender gap might have been larger

Exercise 3: Linear models

Linear Regression

a)

```
library(ISLR2)
```

```
##
## Attaching package: 'ISLR2'

## The following object is masked from 'package:MASS':
##
## Boston
```

```
data(Boston)
```

```
str(Boston)      # 506 rows, 13 columns
```

```
## 'data.frame':  506 obs. of  13 variables:
## $ crim   : num  0.00632 0.02731 0.02729 0.03237 0.06905 ...
## $ zn     : num  18 0 0 0 0 12.5 12.5 12.5 12.5 ...
## $ indus  : num  2.31 7.07 7.07 2.18 2.18 2.18 7.87 7.87 7.87 7.87 ...
## $ chas   : int   0 0 0 0 0 0 0 0 0 0 ...
## $ nox    : num  0.538 0.469 0.469 0.458 0.458 0.458 0.524 0.524 0.524 0.524 ...
## $ rm     : num  6.58 6.42 7.18 7 7.15 ...
```

```
## $ age : num 65.2 78.9 61.1 45.8 54.2 58.7 66.6 96.1 100 85.9 ...
## $ dis : num 4.09 4.97 4.97 6.06 6.06 ...
## $ rad : int 1 2 2 3 3 3 5 5 5 ...
## $ tax : num 296 242 242 222 222 222 311 311 311 311 ...
## $ ptratio: num 15.3 17.8 17.8 18.7 18.7 18.7 15.2 15.2 15.2 15.2 ...
## $ lstat : num 4.98 9.14 4.03 2.94 5.33 ...
## $ medv : num 24 21.6 34.7 33.4 36.2 28.7 22.9 27.1 16.5 18.9 ...
```

```
head(Boston)
```

```
##      crim zn indus chas   nox   rm age   dis rad tax ptratio lstat medv
## 1 0.00632 18  2.31    0 0.538 6.575 65.2 4.0900  1 296   15.3  4.98 24.0
## 2 0.02731  0  7.07    0 0.469 6.421 78.9 4.9671  2 242   17.8  9.14 21.6
## 3 0.02729  0  7.07    0 0.469 7.185 61.1 4.9671  2 242   17.8  4.03 34.7
## 4 0.03237  0  2.18    0 0.458 6.998 45.8 6.0622  3 222   18.7  2.94 33.4
## 5 0.06905  0  2.18    0 0.458 7.147 54.2 6.0622  3 222   18.7  5.33 36.2
## 6 0.02985  0  2.18    0 0.458 6.430 58.7 6.0622  3 222   18.7  5.21 28.7
```

```
summary(Boston)
```

```
##      crim              zn              indus              chas
## Min.   : 0.00632   Min.   : 0.00   Min.   : 0.46   Min.   :0.00000
## 1st Qu.: 0.08205   1st Qu.: 0.00   1st Qu.: 5.19   1st Qu.:0.00000
## Median : 0.25651   Median : 0.00   Median : 9.69   Median :0.00000
## Mean   : 3.61352   Mean   : 11.36   Mean   :11.14   Mean   :0.06917
## 3rd Qu.: 3.67708   3rd Qu.: 12.50   3rd Qu.:18.10   3rd Qu.:0.00000
## Max.   :88.97620   Max.   :100.00   Max.   :27.74   Max.   :1.00000
##      nox              rm              age              dis
## Min.   :0.3850   Min.   :3.561   Min.   : 2.90   Min.   : 1.130
## 1st Qu.:0.4490   1st Qu.:5.886   1st Qu.: 45.02   1st Qu.: 2.100
## Median :0.5380   Median :6.208   Median : 77.50   Median : 3.207
## Mean   :0.5547   Mean   :6.285   Mean   : 68.57   Mean   : 3.795
## 3rd Qu.:0.6240   3rd Qu.:6.623   3rd Qu.: 94.08   3rd Qu.: 5.188
## Max.   :0.8710   Max.   :8.780   Max.   :100.00   Max.   :12.127
##      rad              tax              ptratio              lstat
## Min.   : 1.000   Min.   :187.0   Min.   :12.60   Min.   : 1.73
## 1st Qu.: 4.000   1st Qu.:279.0   1st Qu.:17.40   1st Qu.: 6.95
## Median : 5.000   Median :330.0   Median :19.05   Median :11.36
## Mean   : 9.549   Mean   :408.2   Mean   :18.46   Mean   :12.65
## 3rd Qu.:24.000   3rd Qu.:666.0   3rd Qu.:20.20   3rd Qu.:16.95
## Max.   :24.000   Max.   :711.0   Max.   :22.00   Max.   :37.97
##      medv
## Min.   : 5.00
## 1st Qu.:17.02
## Median :21.20
## Mean   :22.53
## 3rd Qu.:25.00
## Max.   :50.00
```

b)


```
lm.fit <- lm(medv ~ lstat, data = Boston)
summary(lm.fit)
```

```
##
## Call:
## lm(formula = medv ~ lstat, data = Boston)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -15.168  -3.990  -1.318   2.034  24.500
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) 34.55384    0.56263   61.41  <2e-16 ***
## lstat       -0.95005    0.03873  -24.53  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 6.216 on 504 degrees of freedom
## Multiple R-squared:  0.5441, Adjusted R-squared:  0.5432
## F-statistic: 601.6 on 1 and 504 DF,  p-value: < 2.2e-16
```

```
# Alternatively attach Boston first
# attach(Boston)
# lm.fit <- lm(medv ~ lstat)
coef(lm.fit) # to access coefficients only
```

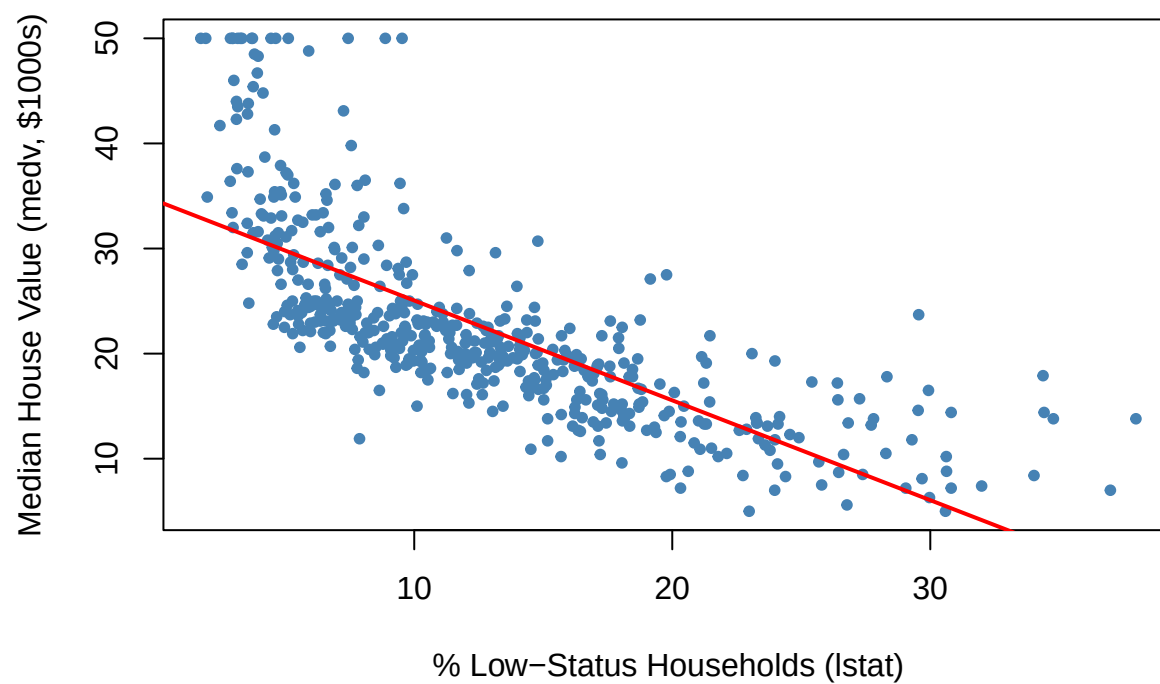
```
## (Intercept)      lstat
## 34.5538409  -0.9500494
```

Intercept $\hat{\beta}_0 = 34.55$: When $\text{lstat} = 0$ (no low-status households), the predicted median house value is \$34,550, on average. Slope $\hat{\beta}_1 = -0.95$: For each 1 percentage point increase in lstat , median house value decreases by approximately \$950, on average. This effect is statistically significant on a 1% level.

c)

```
plot(Boston$lstat, Boston$medv,
     xlab = "% Low-Status Households (lstat)",
     ylab = "Median House Value (medv, $1000s)",
     main = "medv vs. lstat",
     pch = 20, col = "steelblue")
abline(lm.fit, col = "red", lwd = 2)
```

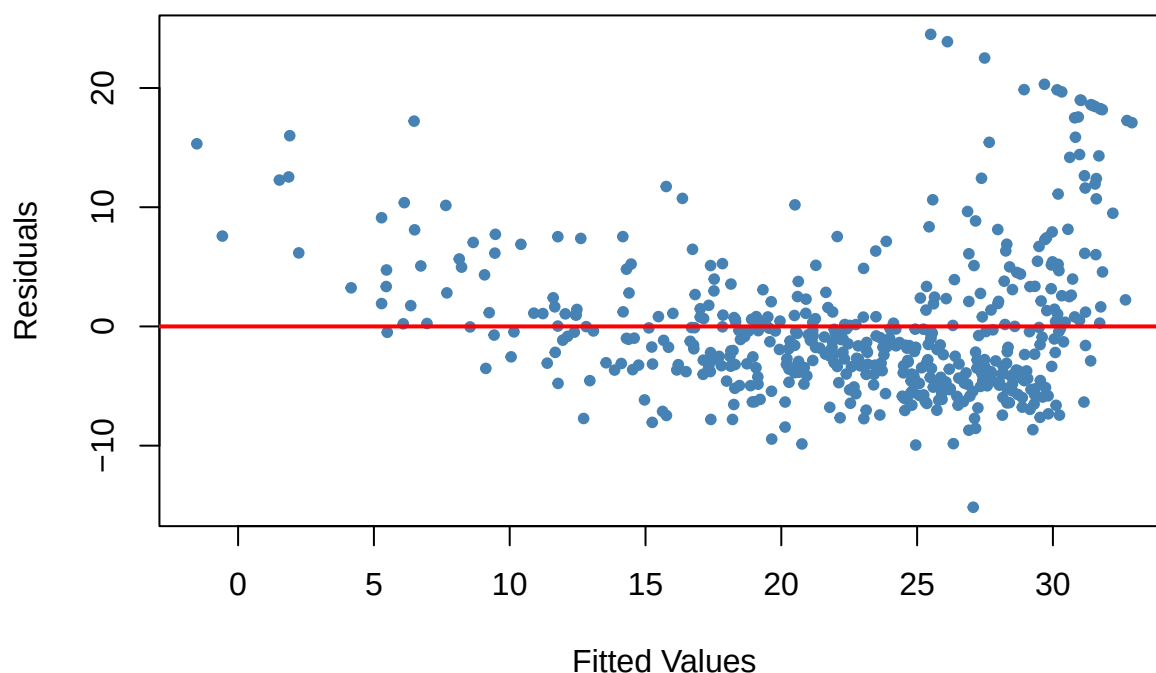
medv vs. lstat



d)

```
plot(lm.fit$fitted.values, lm.fit$residuals,  
     xlab = "Fitted Values",  
     ylab = "Residuals",  
     main = "Residuals vs Fitted Values",  
     pch = 20, col = "steelblue")  
abline(h = 0, col = "red", lwd = 2)
```

Residuals vs Fitted Values



e)

```
# 95 % confidence interval
predict(lm.fit, data.frame(lstat = (c(5, 10, 15))), interval = "confidence")
```

```
##          fit      lwr      upr
## 1 29.80359 29.00741 30.59978
## 2 25.05335 24.47413 25.63256
## 3 20.30310 19.73159 20.87461
```

```
# 95 % prediction interval
predict(lm.fit, data.frame(lstat = (c(5, 10, 15))), interval = "prediction")
```

```
##          fit      lwr      upr
## 1 29.80359 17.565675 42.04151
## 2 25.05335 12.827626 37.27907
## 3 20.30310  8.077742 32.52846
```

CI: a range of values that is likely to contain the *value of an unknown population parameter* with a specified probability
 PI: a range of values that is expected to contain a *single future observation* with a certain probability
 Prediction intervals account for future observation Y_* variability (due to the noise ϵ_*), while confidence intervals only account for uncertainty in estimating the mean.

f)

```
lm.fit2 <- lm(medv ~ ., data = Boston)
summary(lm.fit2)
```

```
##
## Call:
## lm(formula = medv ~ ., data = Boston)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -15.1304  -2.7673  -0.5814   1.9414  26.2526
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  41.617270   4.936039   8.431 3.79e-16 ***
## crim        -0.121389   0.033000  -3.678 0.000261 ***
## zn           0.046963   0.013879   3.384 0.000772 ***
## indus        0.013468   0.062145   0.217 0.828520
## chas         2.839993   0.870007   3.264 0.001173 **
## nox        -18.758022   3.851355  -4.870 1.50e-06 ***
## rm           3.658119   0.420246   8.705 < 2e-16 ***
## age          0.003611   0.013329   0.271 0.786595
## dis        -1.490754   0.201623  -7.394 6.17e-13 ***
## rad          0.289405   0.066908   4.325 1.84e-05 ***
## tax        -0.012682   0.003801  -3.337 0.000912 ***
## ptratio     -0.937533   0.132206  -7.091 4.63e-12 ***
## lstat       -0.552019   0.050659 -10.897 < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 4.798 on 493 degrees of freedom
## Multiple R-squared:  0.7343, Adjusted R-squared:  0.7278
## F-statistic: 113.5 on 12 and 493 DF, p-value: < 2.2e-16
```

```
# Alternatively use lm(y x1 + x2 + x3)
```

Improved prediction as adjusted R^2 increases from 0.54 to 0.73

- g) `medv` is a continuous numeric variable. Classification models like logistic regression or LDA require a categorical response variable. To apply logistic regression or LDA, you would need to discretize `medv` into categories, for example into *low*, *medium* and *high* median house values

Logistic Regression

a)

```
library(ISLR2)
str(Smarket)
```

```
## 'data.frame':  1250 obs. of  9 variables:
## $ Year      : num  2001 2001 2001 2001 2001 ...
## $ Lag1      : num  0.381 0.959 1.032 -0.623 0.614 ...
```

```
## $ Lag2      : num  -0.192 0.381 0.959 1.032 -0.623 ...
## $ Lag3      : num  -2.624 -0.192 0.381 0.959 1.032 ...
## $ Lag4      : num  -1.055 -2.624 -0.192 0.381 0.959 ...
## $ Lag5      : num   5.01 -1.055 -2.624 -0.192 0.381 ...
## $ Volume     : num   1.19 1.3 1.41 1.28 1.21 ...
## $ Today      : num   0.959 1.032 -0.623 0.614 0.213 ...
## $ Direction: Factor w/ 2 levels "Down","Up": 2 2 1 2 2 2 1 2 2 2 ...
```

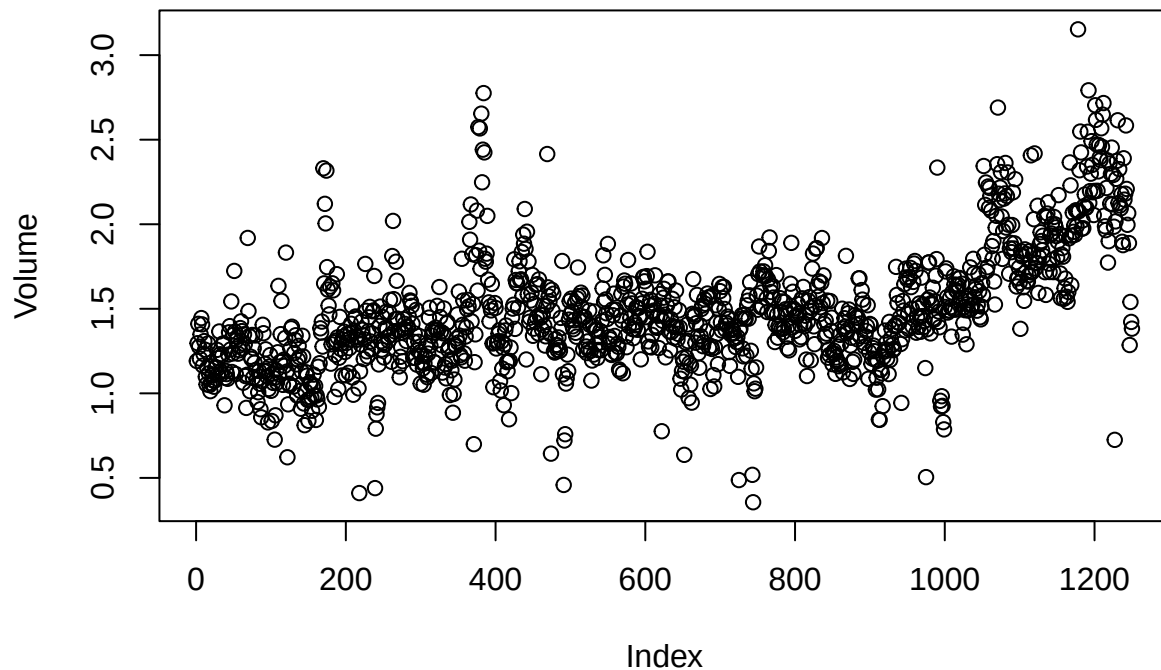
```
summary(Smarket)
```

```
##      Year      Lag1      Lag2      Lag3
## Min.   :2001   Min.   : -4.922000   Min.   : -4.922000   Min.   : -4.922000
## 1st Qu.:2002   1st Qu.: -0.639500   1st Qu.: -0.639500   1st Qu.: -0.640000
## Median :2003   Median : 0.039000   Median : 0.039000   Median : 0.038500
## Mean   :2003   Mean   : 0.003834   Mean   : 0.003919   Mean   : 0.001716
## 3rd Qu.:2004   3rd Qu.: 0.596750   3rd Qu.: 0.596750   3rd Qu.: 0.596750
## Max.   :2005   Max.   : 5.733000   Max.   : 5.733000   Max.   : 5.733000
##      Lag4      Lag5      Volume      Today
## Min.   : -4.922000   Min.   : -4.92200   Min.   : 0.3561   Min.   : -4.922000
## 1st Qu.: -0.640000   1st Qu.: -0.64000   1st Qu.: 1.2574   1st Qu.: -0.639500
## Median : 0.038500   Median : 0.03850   Median : 1.4229   Median : 0.038500
## Mean   : 0.001636   Mean   : 0.00561   Mean   : 1.4783   Mean   : 0.003138
## 3rd Qu.: 0.596750   3rd Qu.: 0.59700   3rd Qu.: 1.6417   3rd Qu.: 0.596750
## Max.   : 5.733000   Max.   : 5.73300   Max.   : 3.1525   Max.   : 5.733000
## Direction
## Down:602
## Up  :648
##
##
##
##
```

```
head(Smarket)
```

```
##   Year  Lag1  Lag2  Lag3  Lag4  Lag5 Volume  Today Direction
## 1 2001  0.381 -0.192 -2.624 -1.055  5.010 1.1913  0.959      Up
## 2 2001  0.959  0.381 -0.192 -2.624 -1.055 1.2965  1.032      Up
## 3 2001  1.032  0.959  0.381 -0.192 -2.624 1.4112 -0.623     Down
## 4 2001 -0.623  1.032  0.959  0.381 -0.192 1.2760  0.614      Up
## 5 2001  0.614 -0.623  1.032  0.959  0.381 1.2057  0.213      Up
## 6 2001  0.213  0.614 -0.623  1.032  0.959 1.3491  1.392      Up
```

```
attach(Smarket)
plot(Volume)
```



b) Linear regression predicts a continuous numeric output. If we code **Direction** as 0 (Down) and 1 (Up) and fit a linear model, two problems arise:

- Predicted values can fall outside $[0, 1]$ - e.g. -0.3 or 1.4 - which has no meaningful interpretation as a probability.
- It assumes the relationship between predictors and the response is linear and continuous, but **Direction** is a binary categorical variable with no natural numeric scale

c)

```
glm.fits <- glm(Direction ~ Lag1 + Lag2 + Lag3 + Lag4 + Lag5 + Volume,
               data = Smarket, family = binomial)

summary(glm.fits)
```

```
##
## Call:
## glm(formula = Direction ~ Lag1 + Lag2 + Lag3 + Lag4 + Lag5 +
##      Volume, family = binomial, data = Smarket)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept) -0.126000   0.240736  -0.523   0.601
## Lag1        -0.073074   0.050167  -1.457   0.145
```

```
## Lag2      -0.042301    0.050086   -0.845    0.398
## Lag3      0.011085    0.049939    0.222    0.824
## Lag4      0.009359    0.049974    0.187    0.851
## Lag5      0.010313    0.049511    0.208    0.835
## Volume    0.135441    0.158360    0.855    0.392
##
## (Dispersion parameter for binomial family taken to be 1)
##
##      Null deviance: 1731.2  on 1249  degrees of freedom
## Residual deviance: 1727.6  on 1243  degrees of freedom
## AIC: 1741.6
##
## Number of Fisher Scoring iterations: 3
```

```
coef(glm.fits)
```

```
## (Intercept)      Lag1      Lag2      Lag3      Lag4      Lag5
## -0.126000257 -0.073073746 -0.042301344  0.011085108  0.009358938  0.010313068
##      Volume
##  0.135440659
```

Lag1 (-0.073): a one-unit increase in yesterday's return is associated with a decrease in the log-odds of the market going Up today. As an odds ratio: $\exp(-0.073) \approx 0.93$ - the odds of Up are multiplied by 0.93, i.e. a $\sim 7\%$ decrease. This makes intuitive sense: if the market rose yesterday, it is slightly less likely to rise again today (mean reversion) Lag2 (-0.042): same direction as Lag1 but smaller effect. Two days ago matters less than yesterday Lag3-Lag5: very small coefficients, close to zero. Returns from 3-5 days ago have negligible predictive value Volume (0.135): higher trading volume is associated with slightly higher odds of an Up day, but the effect is small. None of the effects are statistically significant

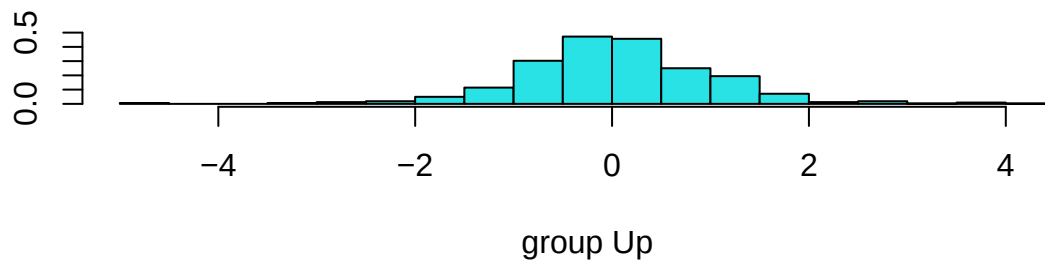
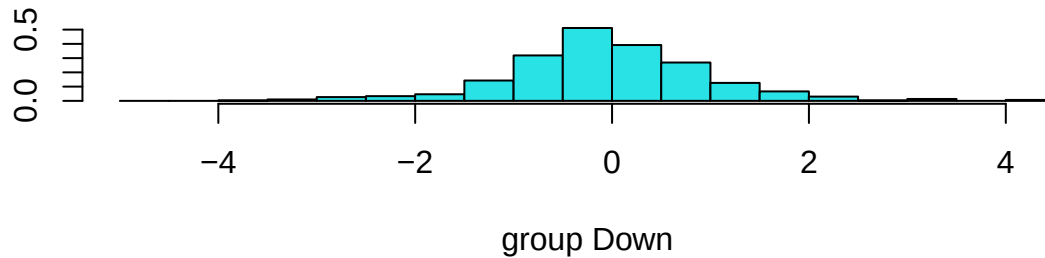
Linear Discriminant Analysis

a)

```
library(MASS)
lda.fit <- lda(Direction ~ Lag1 + Lag2 , data = Smarket)
lda.fit
```

```
## Call:
## lda(Direction ~ Lag1 + Lag2, data = Smarket)
##
## Prior probabilities of groups:
##   Down   Up
## 0.4816 0.5184
##
## Group means:
##           Lag1      Lag2
## Down  0.05068605  0.03229734
## Up   -0.03969136 -0.02244444
##
## Coefficients of linear discriminants:
##           LD1
## Lag1 -0.7567605
## Lag2 -0.4707872
```

```
plot(lda.fit)
```



- b) Prior probabilities reflect the class proportions in the training data. Here they are nearly equal ($\sim 49\%$ Down, $\sim 51\%$ Up), meaning the market went up and down roughly equally often.

Group means show the average value of each predictor within each class: On days followed by a Down move, Lag1 and Lag2 tend to be slightly positive, prior returns were marginally up. On days followed by an Up move, Lag1 and Lag2 tend to be slightly negative, prior returns were marginally down.